

WAZUH SIEM HOME LAB

SIEM | File Integrity Monitoring | Brute Force Detection | MITRE
ATT&CK

SOC Security Operations Manual

This lab manual is designed for educational and home lab purposes only. All attack simulations must be performed exclusively on machines you own or have explicit permission to test. Never perform these activities on production systems, corporate networks, or any external infrastructure.

 pavankumar022

 pavankumar022

Contents

1. Overview & Lab Architecture
2. Prerequisites
3. Phase 1 Wazuh Manager Setup
4. Phase 2 File Integrity Monitoring
5. Phase 3 Attack Simulation from Kali
6. Phase 4 Network Capture & Firewall Evidence
7. Phase 5 Vulnerability Scan
8. Evidence Reference Index
9. Troubleshooting

pavankumar022.pdf

Overview & Lab Architecture

This manual provides a complete, repeatable guide for building a professional Wazuh SIEM home lab. Following this guide produces real security evidence suitable for a cybersecurity portfolio and SOC analyst applications.

What You Will Build

- 3-machine security operations lab with real attack simulation and detection
- Wazuh SIEM with File Integrity Monitoring — file create, modify, delete events
- Brute force attack from Kali Linux, fully detected and alerted by Wazuh
- Automatic MITRE ATT&CK technique mapping (T1110, T1021.004)
- Network traffic capture via Wireshark showing Nmap SYN scan packets
- iptables firewall rules blocking attacker IP with evidence export
- Vulnerability scan report using Nmap vuln scripts
- 15 evidence screenshots and 3 exported files

Lab Architecture

Machine	OS	IP	Role
Ubuntu VM	Ubuntu Server 22.04	192.168.1.10	Wazuh Manager + Dashboard
Windows PC	Windows 11 Home	192.168.1.5	Wazuh Agent (Victim)
Kali Linux	Kali 2026.1	192.168.1.7	Attacker Machine

NOTE: All three machines must be on the same subnet (192.168.1.x). Set VirtualBox network adapter to **Bridged Adapter** for the Ubuntu VM.

Prerequisites

- VirtualBox installed on host machine
- Ubuntu Server 22.04 LTS ISO from ubuntu.com/download/server
- Minimum 8 GB RAM on host, 4 GB allocated to Ubuntu VM
- 50 GB free disk space for the VM
- Home network only — do not use college or public Wi-Fi
- Basic Linux command-line knowledge

IMPORTANT: Use Wazuh agent version 4.12.0 to match the manager. Agent version must be equal to or lower than manager version.

pavankumar022.pdf

Phase 1 — Wazuh Manager Setup

Step 1 — Install Ubuntu Server VM

1. Download Ubuntu Server 22.04 LTS from ubuntu.com/download/server
2. Create new VM in VirtualBox: 4 GB RAM, 2 CPU cores, 50 GB disk
3. Set Network: Settings → Network → Adapter 1 → **Bridged Adapter**
4. Boot from ISO and complete the installation wizard
5. After boot, verify network IP:

```
ip a | grep inet
```

[SS-01] Ubuntu VM Network Configuration

Terminal showing `ip a` output with 192.168.1.x IP confirmed on same subnet as Windows and Kali.

Step 2 — Install Wazuh Manager

WARNING: Takes 10–15 minutes. Save the admin credentials displayed at the end.

```
sudo apt update && sudo apt upgrade -y
curl -sO https://packages.wazuh.com/4.12/wazuh-install.
sh
sudo bash ./wazuh-install.sh -a -i
```

Verify all three services are running:

```
sudo systemctl status wazuh-manager
sudo systemctl status wazuh-indexer
sudo systemctl status wazuh-dashboard
```

If any service shows failed, restart it:

```
sudo systemctl restart wazuh-indexer
sudo systemctl restart wazuh-manager
sudo systemctl restart wazuh-dashboard
```

Step 3 — Access Wazuh Dashboard

On your Windows PC, open a browser and go to:

```
https://192.168.1.10
```

Accept the browser security warning (Advanced → Proceed), then login with username **admin** and the password from the installer.

[SS-02] Wazuh Dashboard Home Page

Browser showing Wazuh dashboard at https://192.168.1.10, 0 agents registered, all module cards visible.

pavankumar022.pdf

Step 4 — Install Wazuh Agent on Windows

Open **PowerShell** as **Administrator** on the Windows PC:

```
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/  
windows/wazuh-agent-4.12.0-1.msi -OutFile $env:tmp\  
wazuh-agent-412.msi
```

```
msiexec.exe /i $env:tmp\wazuh-agent-412.msi /qn  
WAZUH_MANAGER="192.168.1.10" WAZUH_AGENT_NAME="  
WindowsHost"
```

```
Start-Sleep -Seconds 15  
Get-Service WazuhSvc
```

Step 5 — Register Agent with Manager

On the Ubuntu VM, generate the authentication key:

```
sudo /var/ossec/bin/manage_agents
```

1. Press **A** → Add agent
2. Name: WindowsHost
3. IP: type any
4. Confirm with **Y** — note the Agent ID (e.g. 001)
5. Press **E** → Extract key → enter agent ID → copy the full key string

On Windows PowerShell (Admin), apply the key:

```
& "C:\Program Files (x86)\ossec-agent\manage_agents.exe"  
-i [PASTE_KEY_HERE]
```

Edit ossec.conf to fix the manager IP:

```
notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"
```

Find `<address>0.0.0.0</address>` and change to `<address>192.168.1.10</address>`
Save and close.

```
Start-Service WazuhSvc  
Get-Service WazuhSvc
```

[SS-03] Windows Agent Active on Wazuh Dashboard

Wazuh Agents page showing WindowsHost with status Active (1) and green indicator. Agent ID 001.

pavankumar022.pdf

Phase 2 — File Integrity Monitoring

Step 6 — Configure FIM Directory

Create the monitored folder on Windows:

```
mkdir C:\Users\pavan\MonitoredFolder
```

Open the agent configuration file:

```
notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"
```

Inside the <syscheck> section, add this line after <frequency>43200</frequency>:

```
<directories realtime="yes">C:\Users\pavan\
  MonitoredFolder</directories>
```

NOTE: The `realtime="yes"` attribute triggers immediate alerts on file changes instead of waiting for the 12-hour scheduled scan.

Restart the agent to apply changes:

```
Restart-Service WazuhSvc
```

Step 7 — Trigger FIM Events

Wait 2–3 minutes after restart for the baseline scan to complete. Then run each command and wait 30 seconds between them.

Create a file:

```
echo "Sensitive Data" > C:\Users\pavan\MonitoredFolder\
  test.txt
```

[SS-04] FIM Alert — File Added

Wazuh FIM Events: syscheck.event=added, path=test.txt, rule.id=554, rule.level=5, description=File added to the system.

Modify the file:

```
echo "Modified content" >> C:\Users\pavan\
MonitoredFolder\test.txt
```

[SS-05] FIM Alert — File Modified

Wazuh FIM Events: syscheck.event=modified, rule.id=550, rule.level=7, description=Integrity checksum changed.

Delete the file:

```
del C:\Users\pavan\MonitoredFolder\test.txt
```

[SS-06] FIM Alert — File Deleted

Wazuh FIM Events: syscheck.event=deleted, rule.id=553, rule.level=7, description=File deleted.

pavankumar022.pdf

Navigate to: Dashboard → Menu → **File Integrity Monitoring** → **Events**

Phase 3 — Attack Simulation from Kali

Step 8 — Nmap Port Scan

On the Kali Linux terminal:

```
nmap -sV -p 1-1000 192.168.1.5 -oN nmap_scan_report.txt
cat nmap_scan_report.txt
```

[SS-07] Nmap Port Scan Terminal Output

Kali terminal: ports 135/tcp (msrpc) and 445/tcp (microsoft-ds) open on Windows 192.168.1.5.

Step 9 — Enable SSH on Ubuntu

```
sudo apt install openssh-server -y
sudo systemctl enable ssh --now
sudo systemctl status ssh
sudo ss -tlnp | grep 22
```

NOTE: SSH must show **active (running)** and port 22 listening before running Hydra.

Step 10 — Hydra SSH Brute Force Attack

On Kali Linux:

```
hydra -l root -P /usr/share/wordlists/rockyou.txt ssh
: //192.168.1.10 -t 4 -V
```

Let it run 2–3 minutes showing [ATTEMPT] lines, then press **Ctrl+C**.

[SS-09] Hydra Brute Force Terminal

Hydra attacking ssh://192.168.1.10:22. Multiple [ATTEMPT] lines visible. 14,344,399 total tries queued.

Step 11 — View Brute Force Alerts on Wazuh

1. Go to Menu → **Threat Hunting** → **Events**
2. Filter: `manager.name: UBUNTULINUX`
3. Time range: Last 15 minutes → Refresh

[SS-10] Wazuh Brute Force Detection — 2367 Alerts

Threat Hunting showing 2,367 hits, large spike in timeline. syslog: User missed the password more than one time (rule 2502, level 10); sshd: authentication failed (rule 5760, level 5).

Step 12 — View MITRE ATT&CK Mapping

Click on any `sshd: authentication failed` row to expand Document Details.

[SS-11] MITRE ATT&CK Mapping (T1110)

rule.mitre.id=T1110.001 T1021.004 | tactic=Credential Access, Lateral Movement | technique=Password Guessing, SSH | fireddtimes=92 | gdpr=IV_35.7.d | hipaa=164.312.b | nist_800_53=AU.14 AC.7 | pci_dss=10.2.4 10.2.5

[SS-12] Multiple Authentication Failures — Level 10

rule.description=Multiple authentication failures | rule.level=10 | mitre.id=T1110 | tactic=Credential Access | technique=Brute Force | previous_output shows max auth attempts from 192.168.1.7

Phase 4 — Network Capture & Firewall Evidence

Step 13 — Wireshark Network Capture

On Kali Linux:

```
sudo wireshark &
```

1. Select **eth0** interface → click **Start Capture**
2. In a new terminal, run Nmap while capturing:

```
nmap -sV 192.168.1.10 -p 1-1000
```

3. Stop capture after 1 minute
4. Apply display filter:

```
tcp.flags.syn == 1
```

5. Save: File → Save As → **wireshark_capture.pcap**

[SS-13] Wireshark SYN Packet Capture

Filter `tcp.flags.syn==1` applied. Source 192.168.1.7 (Kali) to Destination 192.168.1.10 (Ubuntu). SYN packets to sequential ports visible. 2,683 packets captured.

Step 14 — iptables Firewall Rules

On Ubuntu VM:

```
sudo iptables -A INPUT -s 192.168.1.7 -j DROP
sudo iptables -A OUTPUT -d 192.168.1.7 -j DROP
sudo iptables -L -v -n
sudo iptables -L -v -n > /root/firewall_rules_with_block
.txt
cat /root/firewall_rules_with_block.txt
```

After taking the screenshot, remove the rules:

```
sudo iptables -F
```

[SS-14] iptables Firewall — Blocking Kali IP

Chain INPUT DROP for source 192.168.1.7; Chain OUTPUT DROP for destination 192.168.1.7. Exported to /root/firewall_rules_with_block.txt.

pavankumar022.pdf

Phase 5 — Vulnerability Scan

Step 15 — Nmap Vulnerability Scan

On Kali Linux:

```
nmap --script vuln 192.168.1.5 -oN vuln_scan_windows.txt  
cat vuln_scan_windows.txt
```

This runs 20+ vulnerability detection scripts against the Windows target.

[SS-15] Vulnerability Scan Results — Windows Target

Host 192.168.1.5 up. Ports 135 and 445 open. Host script results: samba-vuln-cve-2012-1182 checked; smb-vuln-ms10-054=false; smb-vuln-ms10-061 attempted. Scan completed in 29.73 seconds. Saved to vuln_scan_windows.txt.

pavankumar022.pdf